Epitalon (AEDG) + Crystagen (EDP-3) + Pinealon (EDR)

Epitalon 5–10 mg, Crystagen 0.5–1 mg, Pinealon ≤0.5 mg daily × 10 days; 1–2 courses/year

Epitalon with Crystagen or Vilon and optional Pinealon is an experimental, speculative protocol proposed to support healthspan, immune regulation, circadian function, and neuronal resilience. Epitalon is the synthetic tetrapeptide AEDG; cellular studies demonstrate telomere and oxidative-stress effects, and animal studies report selected age-related benefits. [1–7] Crystagen (EDP) and Vilon (KE) have distinct immune-regulatory findings, while Pinealon (EDR) has neuronal and cognitive findings in experimental models. [10–15] The classic human regimen used the heterogeneous glandular extracts Epithalamin and Thymalin, whose outcomes cannot be assigned directly to these synthetic peptides. [8,18]

💉 💉 Short-term Epitalon tolerability has been reported under specific clinical conditions. [19,22]

⭐ Epitalon has plausible healthspan applications based on selected animal findings and limited human circadian biomarker results. [3,4,7,20,21]

The efficacy of the complete synthetic peptide protocol is speculative.

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

A short clinical report described no side effects or worsening among Epitalon-treated patients. [19] Selected long-term mouse studies also reported reassuring findings under their exposure conditions. [4,6]

Selected animal outcomes support plausible healthspan applications of Epitalon. [3,4,7] Telomere changes and chromatin effects have been observed in cultured human cells; their contribution to whole-body healthspan remains speculative. [1,9] The complete synthetic combination remains speculative.

How Evidence Scope Was Applied in This Review

Evidence was evaluated here with an adjustment for source quality. Much of the Epitalon evidence comes from a single regional source, including small or methodologically limited studies that have not been independently replicated. When a claim depended primarily on this evidence, its evidence classification was reduced to reflect the additional uncertainty while still recognizing the evidence as informative.

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Extraordinary Claims with Important Caveats

Early studies using Epithalamin and Thymalin, the pineal and thymic extracts, reported improvements in immune, endocrine, metabolic, and functional measures. These findings support possible human benefit from those preparations. [8,18]

Khavinson and Morozov reported 266 older adults across two centers and a separate treatment subgroup. The St. Petersburg trial allocated 94 women to placebo, Thymalin, Epithalamin, or both extracts and used daily IM treatment for 10 days, repeated the following year. Another 20 participants received both extracts annually for six years. The Kyiv study included 152 participants and used five IM administrations spaced 2–3 days apart per course, with courses repeated every 5–6 months for three years. Follow-up lasted six years in St. Petersburg and eight years in Kyiv. [18]

The investigators reported mortality rates approximately 2.0–2.1 times lower with Thymalin, 1.6–1.8 times lower with Epithalamin, and 2.5 times lower with the two-course combination than with controls. The separate 20-person annual-combination group had a reported 4.1-fold difference. [8,18] In the St. Petersburg groups, six-year mortality was 81.8% in controls, 41.7% with Thymalin, 45.8% with Epithalamin, 33.3% with the two-course combination, and 20.0% in the separate annual-combination group. [18]

Both centers described stratified randomization; the St. Petersburg four-arm trial was described as double-blind. Allocation concealment, attrition handling, and prespecified outcome methods were insufficiently detailed. The separate annual-combination group was not described as randomized, and included men as well as women, unlike the main St. Petersburg trial. Small groups and different schedules limit causal confidence in the large mortality differences. These results support possible benefit, rather than a reliable prediction of the magnitude or probability of benefit in other populations. [18]

Crucially, these trials used Epithalamin and Thymalin extracts, not synthetic Epitalon or Crystagen. Applying their outcomes to the synthetic protocol is a speculative extrapolation. [8,18]

Replacing Epithalamin with Epitalon

Epitalon (AEDG) is a defined synthetic tetrapeptide developed within the pineal-peptide research program. [3,19] It is a speculative partial functional analogue of Epithalamin. [18,22]

Research Findings and Mechanistic Proposals for Epitalon

The principal findings and proposed applications include:

- **Telomere modulation:** A 2025 study found increased telomere length, hTERT expression, and telomerase activity in normal human epithelial and fibroblast cells after three weeks of

exposure. Breast-cancer cell lines showed telomere lengthening and increased markers of alternative lengthening of telomeres (ALT) after four days, without significantly increased telomerase activity. These patterns support different telomere-maintenance responses by cell context; the precise causal mechanism remains partly inferred. Human healthspan benefit is speculative. [1]

- **Pineal and circadian regulation:** Epitalon increased evening melatonin and altered cortisol rhythmicity in aged female rhesus monkeys. [3] A small human sublingual study reported increased urinary melatonin-metabolite excretion and altered blood-cell clock-gene expression. [20,21] These observations support plausible circadian effects. An isolated rat-pineal experiment found no significant melatonin effect, indicating context dependence. [23] Improved human sleep is speculative.
- **Antioxidant and mitochondrial protection:** Epitalon reduced oxidative-stress measures and improved scratch-wound closure in high-glucose-exposed ARPE-19 retinal cells. [2] Generalized antioxidant benefit is plausible; broader mitochondrial protection is a mechanistic hypothesis.
- **Chromatin and gene regulation:** Epitalon changed chromatin condensation and ribosomal-gene activity in cultured leukocytes from older donors. [9] Broader regulation of maintenance and stress-response programs is a plausible mechanistic interpretation; rejuvenation of a person from these changes remains speculative.

Selecting a Synthetic Thymic Analogue

Thymalin supplied the thymic component of the original extract-based healthspan regimen. It is a heterogeneous animal thymic extract, so no single synthetic peptide can be assumed to reproduce its complete effect. [18]

Vilon (KE) and Crystagen (EDP) are defined short peptides associated with Khavinson's thymic-peptide research. Their immune effects differ. [10,12,17] Either may reproduce selected functions of the extract, but its replacement is speculative. Thymulin is a separate, naturally occurring zinc-dependent thymic hormone with distinct biology. [16]

Candidate Evidence Profiles

Crystagen (EDP): An experimental aging-spleen study reported B-cell activation, but not renewal of aging spleen cells. [12] Human immune and healthspan applications, and replacement of Thymalin, remain speculative.

Vilon (KE): Experimental studies reported T-helper activation associated with reduced apoptosis, and chromatin changes in cultured leukocytes from older donors. [9,12] Female CBA mice showed improved activity, endurance, and survival, with fewer spontaneous neoplasms. [10,11] Human immune and healthspan applications remain speculative; tumor effects depend on the experimental model. [5]

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Thymulin: The zinc-bound nonapeptide participates in T-cell differentiation and immune regulation. Its activity depends on approximately equimolar zinc binding. [16] It is biologically distinct from the short synthetic peptides discussed here. A role as a separate immune-regulatory adjunct is speculative.

Selection among these peptides is theoretical and context dependent. No comparative study establishes one as the closest functional replacement for Thymalin.

Attribute	*Crystagen*	*Vilon*	*Thymulin*
Identity	Synthetic (EDP). [17]	Synthetic (KE). [10]	Thymic zinc-dependent nonapeptide. [16]
Evidence-supported role	B-cell activation without aged-spleen cell renewal in an experimental model. [12]	T-helper activation; selected mouse survival benefits. [10,12]	Zinc-bound form participates in T-cell differentiation. [16]
Evidence scope	Human immune and healthspan applications are speculative.	Human applications are speculative; tumor effects vary by model. [5,10]	Physiological zinc dependence established; benefit of administered zinc-free material is speculative. [16]
Place in this protocol	Speculative candidate synthetic thymic component	Speculative alternative synthetic thymic component	Separate adjunct, not an interchangeable bioregulator or short-course repair peptide

Thymalin / Thymulin Confusion

Although similarly named, “Thymalin” and “Thymulin” are unrelated substances without functional overlap. Thymalin is an animal thymic extract; thymulin is a defined zinc-dependent nonapeptide. [16,18] Thymulin’s effects do not make it an interchangeable synthetic version of Thymalin.

Including Pinealon

In specific rat models, Pinealon (EDR) improved offspring learning after prenatal hyperhomocysteinemia and retention of a learned task during experimental diabetes. [14,15] Cellular experiments demonstrated reduced ROS accumulation and necrotic death, delayed ERK1/2 activation, and altered cell-cycle behavior; the diabetic-rat study also measured NMDA-receptor subunit gene expression. [13,15] Human cognitive benefits and its addition to the combined protocol remain speculative.

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Epitalon (AEDG) + Crystagen (EDP-3) + Pinealon (EDR)

Epitalon 5–10 mg, Crystagen 0.5–1 mg, Pinealon ≤0.5 mg daily × 10 days; 1–2 courses/year

A Modern Synthetic Protocol

A modern synthetic reinterpretation may combine Epitalon with Crystagen or Vilon and optional Pinealon to address complementary pineal, thymic, immune, and neuro-metabolic pathways. The combined protocol and any additive effects remain speculative.

The 10- or 20-day course format is a speculative adaptation of the extract-based Russian regimen rather than an established schedule for the synthetic peptides. Either Crystagen or Vilon may be used as the thymic component, but comparative selection is speculative. Combining these peptides to produce complementary pineal, immune, and neuro-metabolic effects is speculative.

Dosage & Patterns

The extract trials establish what was administered in those studies, not an equivalent synthetic-peptide regimen. [18] Synthetic Epitalon has also been administered by parabulbar and sublingual routes in short clinical studies. [19–21] The SC schedules below are speculative experimental proposals often discussed in petide communities and wellness clinics. Route equivalence cannot be inferred from nominal peptide mass or from general peptide-transporter models. [17,22]

Component	*Reported human exposure / oral proposal*	*Speculative SC schedule*	*Protocol role*
Epitalon	0.5 mg/day sublingual for 20 days in a biomarker study. [20,21]	5–10 mg daily for 10–20 days, once or twice annually	Primary pineal component
Crystagen	Oral 0.5–1 mg daily for 10 days: speculative proposal.	0.5–2 mg daily for 10 days	Candidate thymic component
Vilon	Oral 100–200 mcg daily for 3–10 days: speculative proposal.	1–5 mg daily for 10 days	Alternative thymic component
Pinealon	Oral 300 mcg daily for 10 days: speculative proposal.	Up to 0.5 mg daily for 10 days in combination; separate higher proposal 1–10 mg daily for 10 days	Optional CNS component

Short daily courses: A 10-day course is the principal experimental pattern discussed; three-day and 20-day variants are also speculative proposals.

Sequential pattern: Epitalon first, followed by Crystagen or Vilon, with optional Pinealon afterward, is a speculative sequence.

Overlapping pattern: Starting the next peptide midway through the preceding course is a speculative alternative.

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Repeating the sequence: Once or twice annually is a speculative adaptation of extract-course patterns. [18]

Administration: The human extract study used IM injections; synthetic Epitalon studies used parabulbar injections or sublingual spray. [18–21] The proposed SC and oral schedules are speculative. POT/LAT transport research provides mechanistic hypotheses, not measured human route-to-route dose conversions for these peptides. [17]

Dose Controversy (**Epitalon**)

The St. Petersburg extract trial administered Epithalamin at 10 mg IM daily for 10 days, repeated annually. Kyiv used a different five-administration schedule. [18] Synthetic Epitalon has a different composition and cannot inherit this dose by name association.

Extract doses describe the mass of a heterogeneous preparation, rather than a quantified amount of purified AEDG. A milligram comparison therefore cannot establish that the extract is weaker or that the synthetic peptide should be given at the same mass. [18,22] No validated mass conversion between Epithalamin and synthetic Epitalon was identified. The extract contains multiple components, and its disposition may differ from the isolated tetrapeptide. [17,18,22]

The 5–10 mg daily synthetic Epitalon schedule is a speculative experimental adaptation of the extract-course approach. The short synthetic clinical studies address different routes and endpoints. [18–21]

The speculative, experimental SC healthspan schedule discussed here is:

- **Epitalon: 5–10 mg SC daily for 10–20 days, once or twice annually**

The sublingual biomarker study used 0.5 mg daily for 20 days, divided into two administrations. Among 75 female night-shift workers, 40 with reduced melatonin production were assigned to peptide or placebo; the other 35 provided an age-normal reference group. [20,21] Small groups, short follow-up, and incomplete blinding and assay reporting limit inference. The two reports may describe overlapping participants and are not counted as independent replication. [20–22]

Dose Variations (**Crystagen**)

An injected Crystagen dose is not established by the cited aging-spleen research. [12] The speculative, experimental SC Crystagen schedule discussed here is:

- **Crystagen: 0.5–1 mg SC daily for 10 days; 2 mg daily is also sometimes discussed**

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Dose Variations (**Pinealon**)

The speculative, experimental SC Pinealon schedules discussed here are:

- **Pinealon: up to 0.5 mg SC daily for 10 days in combination**
- **1.0 mg SC daily for 10 day as a separate, higher-dose proposal is also discussed**

Dose Variations (**Vilon**)

The speculative, experimental SC Vilon schedule discussed here is:

- **Vilon: 1–5 mg SC daily for 10 days**

Benefits

The following sections distinguish observed experimental effects, limited human findings, and extrapolated applications. Human extract outcomes and short synthetic-peptide studies address different treatments. [18–21]

1. Cellular Aging, Telomeres & Genomic Stability

- **Telomere maintenance:** Epitalon increased telomere length and telomerase activity in cultured normal human cells. [1] Preserving replicative capacity through this mechanism is plausible.
- **Antioxidant and cellular protection:** Epitalon reduced ROS and improved wound closure in high-glucose-exposed retinal cells, with changes in antioxidant and fibrosis-related gene expression. [2] Broader mitochondrial protection remains a mechanistic hypothesis.
- **Chromatin and gene regulation:** Epitalon altered chromatin condensation and ribosomal-gene activity in cultured leukocytes from older donors. [9] Restoration of beneficial maintenance or repair programs in an intact person is a speculative extrapolation.
- **Antimutagenic and antitumor effects:** Epitalon reduced bone-marrow chromosome aberrations in female SHR mice without changing their total tumor incidence. [4] It reduced selected mammary-tumor and metastasis outcomes in other mouse models. [5,6] These findings support plausible context-specific protective activity.

2. Circadian, Pineal & Endocrine Regulation

- **Circadian and melatonin regulation:** Epitalon increased evening melatonin in aged female rhesus monkeys. [3] A human sublingual study reported increased urinary melatonin-metabolite excretion and blood-cell clock-gene changes, supporting plausible circadian effects. [20,21] Human sleep improvement and broader endocrine coordination remain speculative.

3. Immune Function & Inflammatory Modulation

- **Crystagen immune regulation:** B-cell activation was reported in an aging-spleen model without stimulation of cell renewal. [12] Human immune benefits remain speculative.
- **Vilon immune regulation:** T-helper activation associated with decreased apoptosis was reported experimentally. [12] Human immune benefit remains speculative.

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4. Healthspan Signals

- **Healthspan applications:** In female SHR mice, Epitalon preserved estrous function and increased late-survivor and maximum lifespan measures, but did not increase mean lifespan. [4] Survival effects in senescence-accelerated mice were mixed, depending on substrain and endpoint. [7] These findings support plausible human healthspan applications.
- **Combined synthetic protocol**: Complementary or additive effects from combining Epitalon with Crystagen or Vilon and optional Pinealon remain speculative.

5. Ocular & Retinal Support

- **Retinal support:** Epitalon improved scratch-wound closure and oxidative-stress measures in a retinal-cell model. [2] A separate clinical report described improved visual and electrophysiological outcomes following a 10-day parabulbar course for retinitis pigmentosa, compared with conventional treatment. Randomization and blinding were not described. This supports possible disease-specific benefit; it does not establish retinal protection from the proposed SC protocol. [19]

6. Neuroprotection, Cognitive Resilience, and Synaptic Support (Pinealon-dominant)

- **Neuronal resilience:** Maternal neurons treated with Pinealon reduced oxidative injury in offspring neurons after experimental prenatal hyperhomocysteinemia. [14] Protection of adult human neurons is a plausible mechanistic extrapolation.
- **Cellular signaling:** ROS, necrotic death, ERK1/2 activation, and cell-cycle effects were measured in cellular experiments. [13] NMDA-receptor subunit gene expression was measured in diabetic-rat hippocampus. [15] A unified genome-regulatory mechanism remains speculative.
- **Cognitive function:** Improved learning n was observed in specific rat models of prenatal hyperhomocysteinemia and diabetes. [14,15] General cognitive benefit in humans remains speculative.

Potential Applications

- **Age-related related decline**
- **Broader immune dysfunction**

Cautions

- **Organ transplantation or immunosuppressive therapy**: Interference with intended immunosuppression is a plausible mechanistic concern for immune-modulating peptides, based on the observed lymphocyte effects. [12,16]
- **Active malignancy:** Tumor context matters. Opposite Vilon outcomes across mouse models and Epitalon's effects on telomere maintenance require particular attention. [1,5,10]

Side Effects

Minimal side effects are documented for these peptides. Fatigue and headache are reported, but attribution to the individual peptides have not been characterized.

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Cancer Considerations

Epitalon: Selected animal studies reported lower tumor burden or metastasis, while the SHR study found unchanged total tumor incidence. [4–6] In HER-2/neu mice, lung metastasis size decreased, but the number did not. [5] In vitro telomere lengthening and increased ALT markers in breast-cancer cells raise a mechanistic concern about supporting an existing malignant cell population; they do not demonstrate increased human cancer incidence. [1]

Crystagen: Cancer benefit is speculative. Its experimental B-cell activation does not itself establish favorable tumor surveillance or safety during active malignancy. [12]

Vilon: Female CBA mice showed fewer spontaneous neoplasms, whereas HER-2/neu mice developed more mammary tumors with shorter latency. [5,10,11] The opposing findings support model-dependent effects and prevent a confident low-risk inference for an existing human malignancy.

Clean Window

A speculative “clean window” minimizes concurrent experimental peptides or major protocol changes before, during, and shortly after the course.

Biological Mechanism

The underlying molecular mechanisms of peptide bioregulators such as Epitalon, Vilon, Crystagen, and Pinealon remain poorly characterized. Unlike conventional receptor-targeted peptide drugs, these compounds have no well-established receptor-agonist mechanism, although receptor-mediated activity has not been excluded. Modulation of gene expression is a leading mechanistic hypothesis for how they work, potentially involving direct interactions with DNA, histones, or chromatin that alter gene transcription. Experimental studies provide some support for these mechanisms, but the precise molecular targets and causal pathways remain largely unverified. The scientific literature more often describes these peptides through their observed downstream cellular and physiological effects rather than through well-established mechanisms explaining how those effects originate.

Epitalon: Observed findings include aged-monkey neuroendocrine changes, limited human circadian biomarkers, and cultured-cell telomere, oxidative-stress, and chromatin effects. [1–3,9,20,21]

Crystagen: B-cell activation was reported in an aging-spleen model without accompanying cell renewal. [12]

Vilon: T-helper activation associated with less apoptosis and cultured-leukocyte chromatin changes have been reported. [9,12]

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Pinealon: Reduced ROS accumulation and necrotic death, delayed ERK1/2 activation, and cell-cycle changes were observed in cellular experiments. [13] Neuronal resilience and learning effects were observed in specific rat models. [14,15] Direct genome interaction was proposed by the cell-study authors, but not established by experiments. [13]

Low Dose Epitalon for Sleep

Low dose Epitalon for sleep is a separate application and protocol. Aged-monkey findings and the small sublingual human biomarker study support a plausible circadian rationale for Epitalon. [3,20,21] The human study measured melatonin-metabolite excretion and blood-cell genes, rather than sleep duration, continuity, or insomnia response. Improved sleep from bedtime SC or intranasal administration remains speculative.

The speculative, experimental bedtime dose discussed here is:

- **Epitalon: 50–300 mcg SC or intranasally before bed, continuously or in cycles, co administered with DSIP is also discussed**

Waning perceived effects, cycling requirements, and additive sleep effects from combining Epitalon with DSIP remain speculative. Adaptive changes are a mechanistic possibility; the biomarker studies do not define an optimal continuous or cyclic bedtime schedule. [20,21]

Conclusions

Epitalon has plausible healthspan and circadian applications supported by selected animal findings and limited human biomarker results. [3,4,7,20,21] The combination with Crystagen or Vilon and optional Pinealon remains a speculative reconstruction of the pineal–thymic approach. Safety conclusions depend on the individual endpoint, preparation, route, and duration; short ophthalmic tolerability and reassuring animal observations cannot establish the safety of repeated SC combinations. [4–6,10,19,22]

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